ULTRAPLAST CHAINBELTS PVT. LTD.

Smart Conveyor Planner & Estimator

Complete guide: why we use it and how to use it

For customers, the sales team and the engineering team

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Version 1.0 | 21 October 2026 | Preliminary planning tool - not an engineering design, not a quotation, no prices.

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Part A Why we use this system

The **Ultraplast Smart Conveyor Planner & Estimator** is a free web tool that turns a few simple answers from a customer into a **3D picture, engineering-style drawings, a report and a ready enquiry**. This part explains the problem it solves and what everyone gains.

A1 The problem we are solving

Every conveyor project starts the same way: a customer says "*I need a conveyor*". To give a sensible answer, Ultraplast needs the product size and weight, how many products per minute, the route, the heights and the space available. Customers rarely know what to send, and the information arrives in pieces.

- **Slow first reply.** Sales spend days on phone calls and e-mails just to collect basic data before any real advice can start.
- **Incomplete enquiries.** Missing product size, no throughput, no route - the team cannot judge whether the job is even possible.
- **Customers cannot picture the result.** A list of numbers does not show how the conveyor will look, whether it fits the floor, or how it turns a corner.
- **Problems found late.** A bend that is too tight for the belt, a speed that makes bottles fall over, a route that hits a pillar - these are discovered after time has been spent.
- **Inconsistent information.** Each enquiry arrives in a different format, so engineers must re-type and re-check everything.

A2 What the system does about it

1. **Asks the right questions in a clear order** - four short steps, each with examples, units and a note of what is really needed. Only the conveyor length is compulsory.
2. **Checks the answers immediately** - catches impossible numbers, routes that do not fit the floor, curves that are too tight for the chosen belt, and capacity that is too low.
3. **Shows the result as it is entered** - a live 3D conveyor appears and starts running, so the customer sees what they are describing.
4. **Produces useful documents** - a PDF report, a JSON project file, a drawing sheet and 2D/3D DXF files that open in AutoCAD.
5. **Tells the truth about what is missing** - anything that cannot be calculated safely is marked "Engineering review required" instead of being guessed.
6. **Delivers a structured enquiry** to Ultraplast, so the team starts from complete, comparable information.

A3 Who benefits and how

Who	What they gain
Customer / buyer	Knows exactly what to prepare. Sees the conveyor in 3D within minutes. Learns early if the route does not fit, a curve is too tight, or the speed is too high for the product. Gets a downloadable summary and drawing. No login, no cost. Nothing is sent until they choose to send it.
Sales team	Complete, consistent enquiries. Far fewer back-and-forth calls. A picture and a drawing to discuss on the first call. A plain-language summary of what the customer wants and what is still missing. Optional internal cost estimate (staff only).
Engineering team	A correct starting geometry (route, lengths, curve radii, rise, heights, belt width, product size). 2D and 3D DXF files to open in AutoCAD and continue from. A ready list of inputs still missing. Clear flags for everything needing engineering judgement.
Management	Faster response to enquiries, available any time. A consistent, professional first impression of the company. Better-qualified leads. Very low running cost (a static website with no server to maintain). All product data edited in one place.

A4 Before and after

	Without the tool	With the tool
First customer contact	Vague e-mail or call: "need a conveyor".	Customer fills a guided form and sends a structured enquiry with a drawing.
Collecting data	Several rounds of calls and mails over days.	Done in about 2-10 minutes by the customer; missing items listed automatically.
Seeing the idea	Engineer sketches later; customer imagines.	3D running model and drawings produced instantly.
Finding problems	Discovered during design, or worse, after ordering.	Flagged during entry (tight curve, no fit, speed too high, wrong belt type).
Engineer's start	Re-type numbers from mails.	Open the DXF / JSON; geometry already correct.

A5 The rules the system never breaks

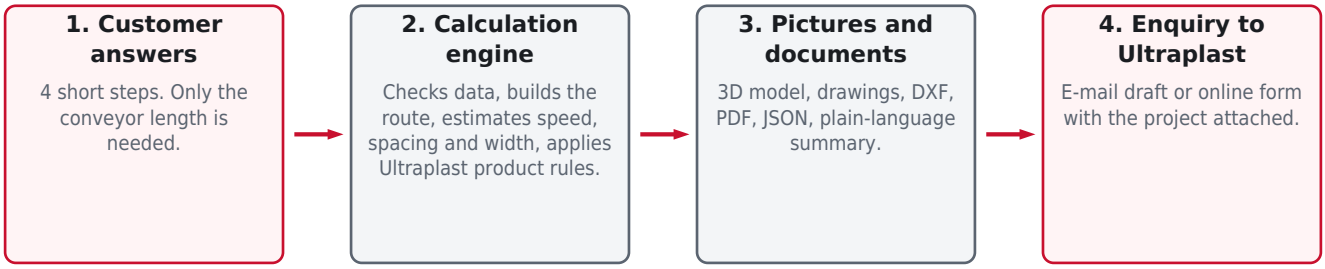
Trust matters more than looking clever. These rules are built into the tool and are checked by automatic tests.

- **No invented prices, capacities or specifications.** Anything that is not entered by the customer or stored in the product data file shows "Engineering review required".
- **Floor area is never treated as conveyor length.** A final design cannot be computed from floor size alone.
- **It never says "Enquiry sent" unless a real service confirmed it.** Otherwise it says clearly that nothing was sent and offers an e-mail draft.
- **No customer data is stored in the browser.** Closing the page clears everything (only the optional staff rate card can be remembered on a staff computer).
- **No secrets in the website code.** Only public web addresses are configured.
- **Honest colours:** black = entered by the customer, red = calculated, grey = indicative only.
- **Every output is marked preliminary** and drawings say "not for manufacture".

> **What this tool is NOT**
>
> It is not a final engineering design, not a quotation, and it shows no prices. It does not select the motor, check belt strength or tension, or check the frame structure. The drawings are indicative and are **not manufacturing drawings**. Ultraplast engineers must review and approve everything before anything is built.

Part B How it works at a glance

The tool is a single web page. Everything runs inside the visitor's own browser, so it is fast, works with a weak connection once loaded, and does not send data anywhere until the customer presses Submit.



What is produced

Output	What it is	Who uses it
Live 3D preview	Conveyor drawn to scale while the form is filled; starts running when enough is known.	Customer
In simple words card	Plain summary: what you told us, what we suggest, what is still needed, what to do now.	Customer, sales
3D model (results)	Rotate, zoom, pan; dimensions; running belt and products; speed and behaviour test.	Customer, sales, engineers
Engineering drawing sheet	A3 sheet: plan, side elevation, end view, floor layout (if given), 3D view, title block.	Sales, engineers
2D DXF / 3D DXF	Full-size (mm) CAD files that open in AutoCAD and similar programs.	Engineers
PDF report	Preliminary project report with tables, schematic, dimension schedule and missing inputs.	Customer, sales
JSON project file	All entries and results in one file; can be re-opened later with Load JSON.	Sales, engineers
Enquiry	Contact details plus the project, sent online or as an e-mail draft.	Sales
Cost estimate	Internal rate-card calculator. Hidden from customers; ships with no prices.	Staff only

Part C How to use it, step by step

C1 Before you start

You do **not** need everything. The tool works with only the conveyor length. But the more you can give, the better the result. If you can, keep these ready:

- The **conveyor length** (the only thing really needed) - or the route: lengths, curve angles and radii, any rise or fall.
- Product **length, width, height** (mm) and **weight**.
- How many products per minute or hour must pass the end.
- Free floor **length and width** (metres) - optional.
- Heights of the belt where products get on and off (mm) - optional.
- Obstacles such as pillars, machines, doors - optional.

> **Not sure? Leave it blank.**
>
> Every box except the conveyor length is optional. Leave anything you do not know empty, or press **Skip this step**. The tool will list it under **Missing engineering inputs** and Ultraplast will confirm it with you.

C2 Two ways to start

- **Quick start** - press **"Quick start: I only know the conveyor length"** on the welcome card. You jump straight to the last step, type the length, and press **Calculate results**.
- **Step by step** - fill the four steps in order. Use **Skip this step** on any page where you do not know the answers.
- **Fill an example for me** - loads sample values (a 12 m bottle line) so you can see how everything looks, then change anything you like.

Look for the small tags next to each box: **NEEDED** (must be filled - only the conveyor length), **HELPFUL** (product size, weight and throughput - improves the result a lot) and **OPTIONAL**. A line at the top of each page says what is needed on that page, and the heading shows "Step 1 of 4".

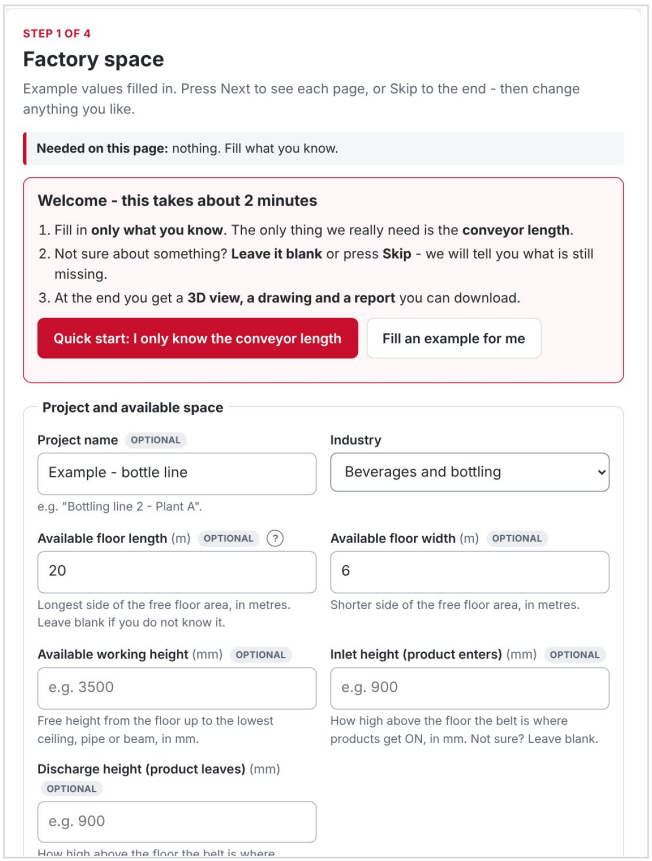


Figure 1 - Step 1. The welcome card, the "Needed on this page" line, tags on every box and plain hints with examples.

C3 Step 1 - Factory space

Describes where the conveyor will go. **Nothing on this page is required.**

Box	What to enter	Example
Project name	A name to recognise the project.	Bottling line 2 - Plant A
Industry	Pick the closest industry (helps suggestions).	Beverages
Floor length / width (m)	The free floor area where the conveyor may be installed. Leave blank if not known - the conveyor is then drawn on its own and the tool does not check the fit.	20 and 6
Working height (mm)	Free height from floor to the lowest ceiling, pipe or beam.	3500
Inlet height (mm)	How high above the floor the belt is where products get ON. If blank, the drawing uses an illustrative 900 mm and says so.	900
Discharge height (mm)	How high the belt is where products get OFF. Differs from the inlet if the belt rises or falls.	1200
Layout	Choose a picture: Straight, L, U, S, Inclined, Curved, Multi-conveyor, Custom. Straight is the default.	Straight
Start / end points	Where the belt starts, measured from the bottom-left corner of the floor plan (X along the length, Y along the width). If blank the route is placed automatically and marked "proposed".	X 0.5, Y 0.5
Obstacles	Add columns, machines or doors with position (X, Y) and size. The tool warns if the route hits one.	Pillar 0.4 x 0.4 m
Drawings	Tick if you have a floor plan; attach it to the enquiry (upload if enabled, otherwise e-mail it).	-

C4 Step 2 - Product

Describes what the conveyor carries. **All optional, but size and weight are very helpful** - without them the tool cannot work out spacing or draw the products.

Box	What to enter	Example
Product name / category	A short name; pick the nearest category (bottles, cartons, trays...). Used to suggest a suitable conveyor type.	500 ml PET bottle
Length / width / height (mm)	Rough size of ONE product. Length is the longest side.	70 / 70 / 200
Product colour	Pick a colour swatch - only changes how the product is drawn in 3D.	Red
Weight per product	Weight of ONE product; choose g or kg in the box next to it.	520 g
Maximum weight, fragility, temperature	Heaviest product expected; how delicate; temperature at the conveyor (blank = normal room temperature).	Moderately fragile
Conditions	Tick Dry, Wet, Oily/greasy, Sticky or Abrasive. Wet and oily lower the grip used in the behaviour check.	Wet
Food contact, cleaning	Hygiene needs and how the belt is cleaned.	Packaged food; washdown
Orientation, gap, accumulation	Which side leads; space between products in mm; whether products queue and touch.	Gap 50 mm

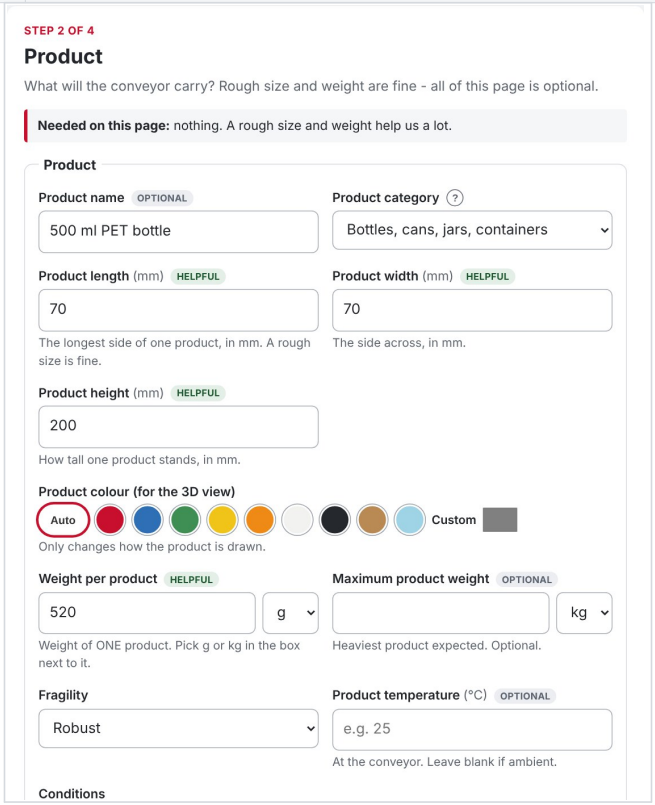


Figure 2 - Step 2. Sizes in millimetres, weight with a unit box, colour swatches for the 3D view.

C5 Step 3 - Production

Describes how much and how fast. **All optional.** If you give a throughput, the tool works out the minimum belt speed.

Box	What to enter	Example
Required throughput	How many products must reach the end. Choose per minute or per hour in the box beside it.	60 per minute
Conveying speed (m/min)	Belt speed if you already know it. If blank, the tool calculates the minimum speed needed.	10

Box	What to enter	Example
Hours per day, shifts, operation	Working pattern; continuous or intermittent.	16 hours; continuous
Feeding, discharge	Manual or automated at the start and end.	Automated
Upstream / downstream machine	Equipment before and after the conveyor.	Filler / packer
Electrical supply	Voltage, phase, frequency - or type "none" if not known.	415 V, 3-phase, 50 Hz

C6 Step 4 - Conveyor (the only step that needs something)

Here you give the **conveyor length** or build the route. This is the one **NEEDED** item. Everything else on the page is optional.

Box	What to enter	Example
Conveyor category / material	Leave on "Automatic recommendation" and "Not sure - recommend" and the tool suggests a type. Or choose one - the tool will warn if it does not suit your route.	Automatic
Conveyor width (mm)	Leave blank for Ultraplast to recommend. If blank, the drawing uses an illustrative width and says so.	200
Belt colour	Pick a swatch or "Auto". Only changes the drawing.	Blue
Belt-to-product friction	Optional. Type a measured value to test your own belt in the behaviour check. Blank = generic value for the belt type.	0.3
Lanes, transfers, infeed, discharge	Products side by side (usually 1); conveyor-to-conveyor transfers (usually 0); where products go on and off (usually 1).	1 / 0 / 1 / 1
Route	Straight layout: just type the **total conveyor length**. Other layouts: fill the segment rows (see C7).	12 m
Washdown, guarding, side guides	Cleaning and safety requirements.	Basic guarding
Advanced engineering inputs	For technical users: pitch, sprocket teeth, pulley diameter - allows a belt-loop length estimate on a simple straight layout.	-

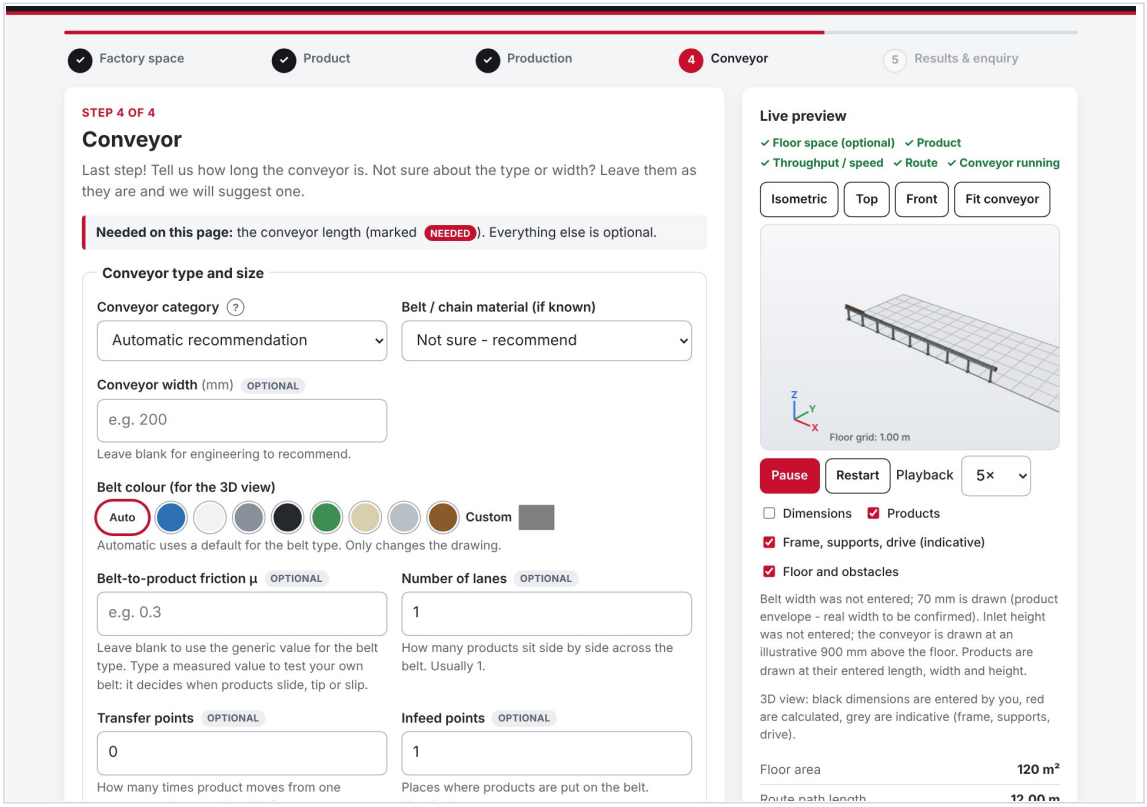


Figure 3 - Step 4. The live 3D preview on the right shows the conveyor running as you complete the form.

C7 Entering a route with curves or height changes

Layout	Rows to fill
Straight	One number: total length (for example 12 m).
L-shaped	3 rows: straight, curve, straight.
U-shaped	5 rows: straight, curve, straight, curve, straight.
S-shaped	5 rows: straight, left curve, straight, right curve, straight.
Inclined	3 rows: straight, incline, straight.
Curved	3 rows: straight, curve, straight.
Custom	Add rows with + Straight, + Curve, + Incline in any order.

- **Straight row** - length in metres.
- **Curve row** - angle (for example 90), **centre-line radius** (for example 800 mm) and Left or Right.
- **Incline row** - horizontal run (m) and either the rise in mm (negative = decline) or the angle.
- The 3D view appears only when every row is complete; a yellow note tells you which row is missing.
- **Floor area is not conveyor length.** Only the route you enter counts.

C8 Reading the results page

After **Calculate results** the results page opens. From top to bottom it contains:

Card	What it tells you
In simple words	The plain-language summary: what you told us, what we suggest, what is still needed, what to do now.
Checks and flags	Red = a problem to fix, yellow = something to check, grey = a note.
Recommended conveyor type	The suggested type with reasons and alternatives. You may choose another in Step 4; the tool then flags any problem.

Card	What it tells you
Your conveyor in 3D	The interactive 3D model (see C9) with the product behaviour check (see C10).
Engineering drawing (CAD)	The drawing sheet and the DXF / SVG downloads (see C11).
Dimension schedule	Table of every segment: length, radius, sweep angle, rise, heights - the same numbers as drawn.
Space and route; Product and throughput; Belt and width	All calculated values, each labelled Customer-provided, Calculated estimate, Configuration-dependent or Engineering review required.
Engineering review required	Items that cannot be safely calculated here: motor and drive, belt tension, structure, sprockets, rollers, return system.
Preliminary equipment checklist	Suggested items only. It is not a bill of materials.
Missing inputs, assumptions, exclusions, next steps	What is still needed, what was assumed, what is not covered, and what to do next.
Download your project summary; Request a quotation	PDF and JSON downloads; the enquiry form (see C12).

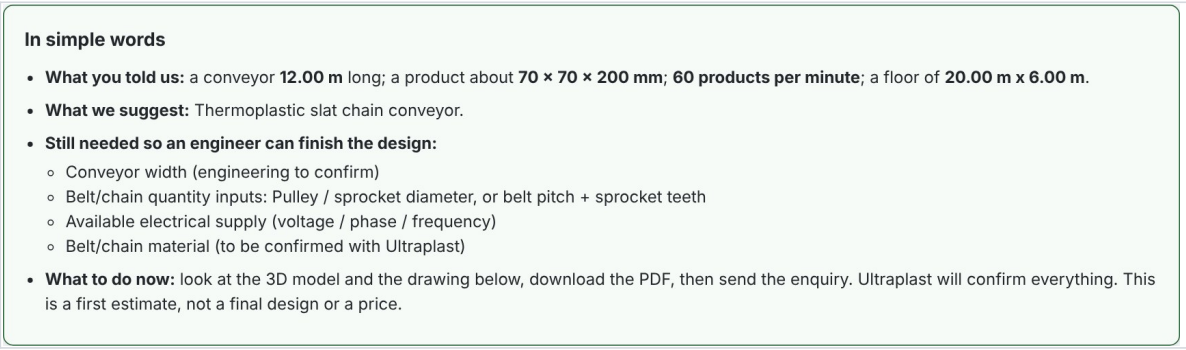


Figure 4 - The "In simple words" card at the top of the results.

C9 The 3D model - how to move it

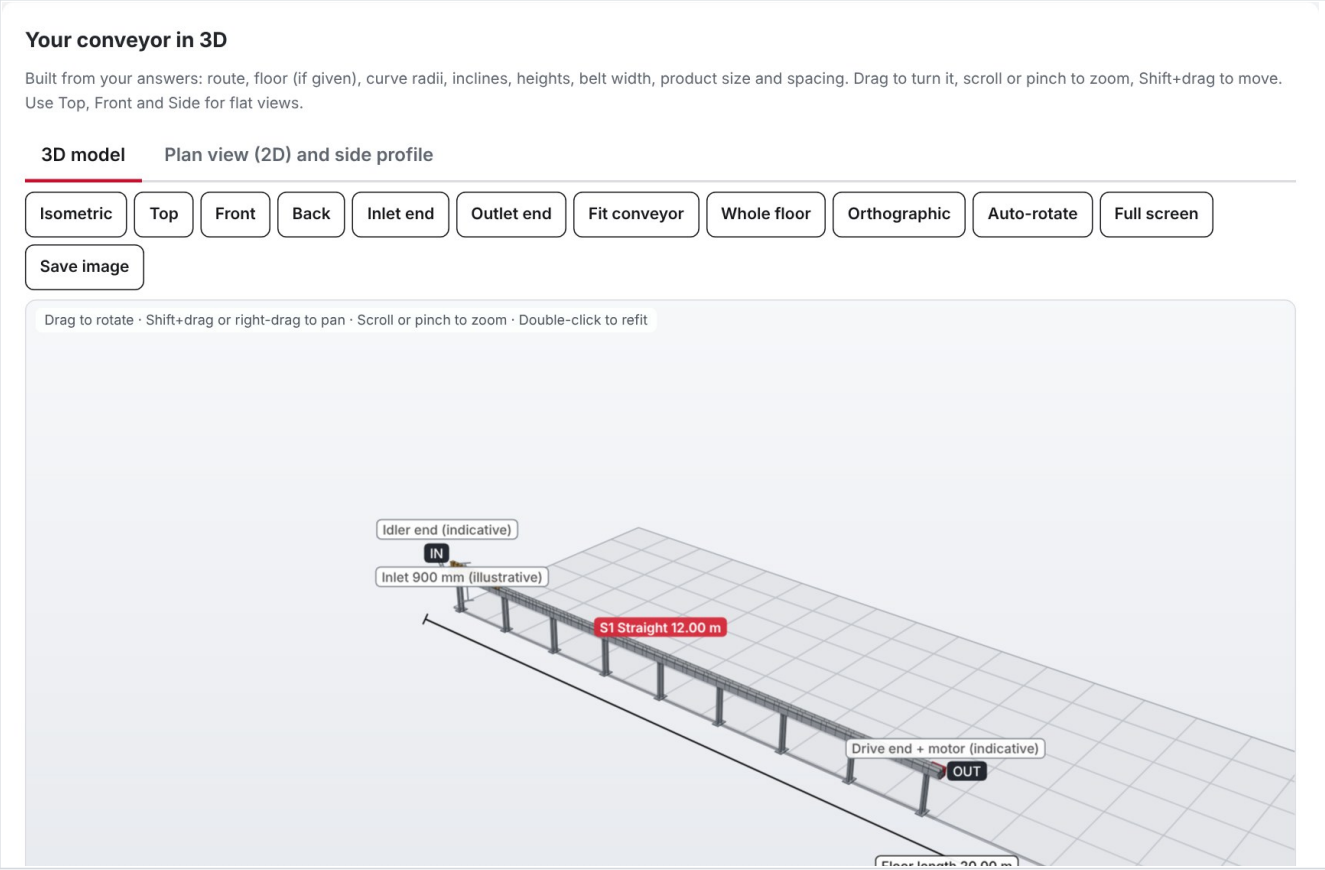


Figure 5 - The 3D model with view buttons, belt speed box, product behaviour check and live counters.

To do this	Do this
Turn the model	Drag with the mouse or one finger.
Zoom	Scroll wheel, or pinch with two fingers.
Move (pan)	Shift + drag, right-drag, or two fingers.
Re-fit to the screen	Double-click, or press Fit conveyor.
Ready-made views	Isometric, Top, Front, Back, Inlet end, Outlet end, Whole floor. (Front / Back / ends follow the direction your conveyor runs.)
True-proportion flat views	Orthographic.
Other buttons	Auto-rotate, Full screen, Save image (PNG picture of the current view).
Show / hide	Tick boxes: Dimensions, Products, Frame/supports/drive, Floor and obstacles. Switch Dimensions off for a cleaner picture.
Run the belt	Pause, Restart and Playback speed. Counters show products on the conveyor and delivered.
Dimension colours	Black = you entered it; red = calculated; grey = indicative (not yet specified by Ultraplast).
Segment tags S1, S2...	Red tags mark the route segments; the Dimension schedule table lists each one.

Tab **"Plan view (2D) and side profile"** keeps the flat top view with the running belt and a side profile for routes that change height.

C10 Belt colour, product colour and "what if the speed is too high?"

The model shows how products behave, so you can see **why** a speed or belt is not suitable. The behaviour depends on the belt type, the product and the speed.

1. Pick a **product colour** (Step 2) and a **belt colour** (Step 4) to see your own colours in the 3D model.
2. In the 3D model, type a higher **Belt speed** in the speed box. If the speed is above the "highest safe speed" shown, products start to **slide or tip off the curve and fall to the floor**. Lower the speed again and they stay on.
3. Press **Stop belt (test)** to see whether tall or slippery products slide or tip when the belt stops.
4. Change the **friction**, **centre-of-gravity height** or **belt stop time** boxes to test your own conditions.

The panel **Product behaviour check** lists each check with a green tick, grey info or red cross: curve (slide and tip), incline (slip), infeed (placing on the moving belt), belt stop, and the discharge end (where products land if nothing receives them). The counters show how many products are on the belt, delivered, or fell off.

Slow belt: products stay on	Fast belt: products slide off the curve	High-grip belt: products tip over

Figure 6 - The same 90-degree curve at 20 m/min (left), at 150 m/min (centre) and with a very high-grip belt at 150 m/min (right). Example values; the numbers in your project will differ.

> **Important - this is a visual aid, not a guarantee**
>
> The friction values and centre-of-gravity ratio are **generic placeholders, not tested by Ultraplast** (the page says so). The model does not include collisions between products, vibration, belt sag, wet or dirty conditions, or real transfers. Use it to discuss limits; Ultraplast engineering confirms the real values.

C11 The engineering drawing (AutoCAD style)

The **Engineering drawing (CAD)** card builds an A3 drawing sheet from your entries and offers three downloads.

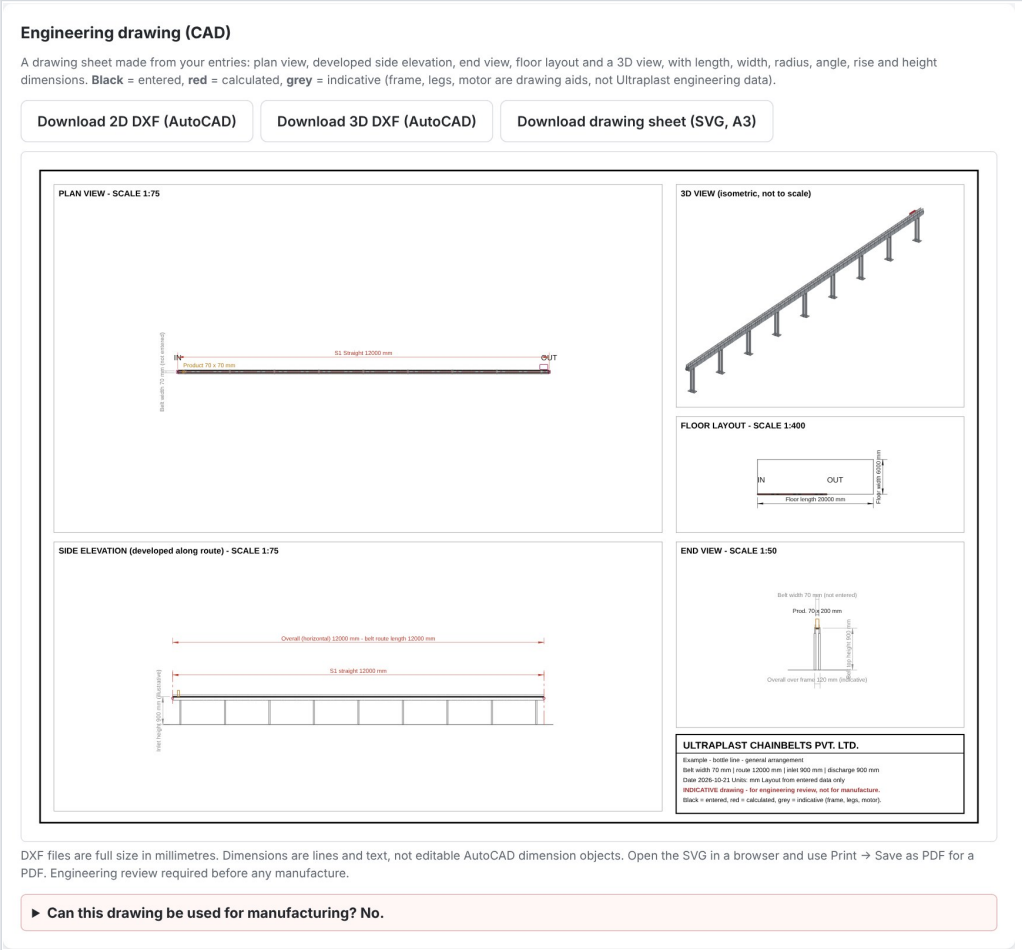


Figure 7 - The drawing sheet: plan view, 3D view, floor layout, side elevation, end view and title block.

Download	What you get	How to use it
2D DXF	Full size in millimetres: plan view, developed side elevation, end view and floor layout, with dimensions (length, width, radius, angle, rise, heights).	In AutoCAD choose OPEN (or INSERT) and select the .dxf file. Type ZOOM then E to see everything.
3D DXF	Every solid face as a 3D face on its own layer (belt, frame, supports, drive, floor), a 3D centre-line and dimension text. The belt takes the nearest AutoCAD colour to your chosen belt colour.	Open the same way, then use the 3D orbit / visual styles.
SVG sheet (A3)	The drawing sheet with scaled views and a title block.	Open in a web browser, then Print and choose Save as PDF (A3 or A4).

- **Colours of dimensions** follow the same code as the 3D view. In the DXF they are on separate layers - `DIM_ENTERED` (black), `DIM_CALC` (red), `DIM_IND` (grey) - so they can be switched on or off.
- Layers whose name ends in `_IND` (frame, guides, supports, drive) are **indicative drawing aids**, not Ultraplast engineering data.
- Dimensions are lines and text, **not** editable AutoCAD dimension objects.
- If the floor size was not entered, the drawing shows the conveyor on its own and the floor-layout view is left out.

Can the drawing be used for manufacturing? No.

It is **indicative and for engineering review**. It has no motor, tension or structural engineering, no bill of materials and no manufacturing details. It is a correct starting point that Ultraplast engineers complete, check and approve. See Part G.

C12 Saving, reports and sending an enquiry

- **Download PDF report** and **Download JSON summary** are on the results page.
- **Load JSON** (top right) re-opens a saved project. **Nothing is saved automatically** - closing the page clears everything, which also protects your privacy.
- **Metric / Imperial** (top right) converts your entries to the other unit system.
- **Advanced** (top right) adds engineering inputs for a belt-loop estimate on a simple straight layout.
- **Request a detailed quotation:** fill your name, company, e-mail, mobile, city, state and timeline, tick the consent box and press **Submit enquiry**. The page says "received" only if the service really confirms it. If online sending is not connected it says so plainly - use **E-mail Ultraplast (draft)** or call, and attach the PDF.
- **Edit inputs** takes you back to change answers; **Start new project** or **Reset** clears everything.

Part D Reading and trusting the results

D1 Colour code

Colour	Meaning	Where
Black	You entered it.	3D labels, drawings, DXF layer DIM_ENTERED
Red	Calculated from your entries (route length, curve arc, rise, heights).	3D labels, drawings, DXF layer DIM_CALC
Grey	Indicative only: frame, guide rails, legs, rollers, motor and similar parts not yet specified by Ultraplast.	3D labels, drawings, DXF layer DIM_IND

D2 Status labels on results

Label	Meaning
Customer-provided	You entered this value.
Calculated estimate	Worked out from your data using documented formulas.
Configuration-dependent	Depends on product data that Ultraplast must confirm.
Engineering review required	Cannot be safely calculated here. An engineer must decide.

D3 Common red problems and how to fix them

Message	What to do
Route does not fit inside the floor	Shorten the route, change the layout, or move the start point. (Only checked when a floor size is entered.)
Route passes through an obstacle	Move the start point or change the layout.
Curve radius is below the published minimum	Increase the radius.
Capacity below required throughput	Increase the speed, add lanes, or reduce the gap between products.
Conveyor narrower than the product	Increase the conveyor width.
Unsupported configuration (for example a straight belt on a curve)	Choose a curve-capable conveyor type, or leave the category on Automatic.

D4 What the tool calculates (in plain words)

- **Route length** - straight lengths added to curve arcs (radius x angle) and incline lengths (the slope length from run and rise).
- **Product pitch** - product length along the belt plus the gap. **Required speed** - products per minute x pitch.
- **Footprint** - the space the route occupies, checked against the floor (only if a floor is entered).
- **Width envelope** - the width the product needs; the real conveyor width is never invented.
- **Checks** - minimum curve radius from published limits, capacity, obstacles, end-point, and heights.

Part E Worked examples

Example 1 - A 12 m bottle line (press "Fill an example for me")

Entry	Value
Floor	20 m x 6 m
Product	500 ml PET bottle, 70 x 70 x 200 mm, 520 g, gap 50 mm
Throughput	60 per minute (speed left blank)
Layout / route	Straight, 12 m

What happens: the product pitch is 70 + 50 = **120 mm**. To deliver 60 per minute the belt needs 60 x 0.120 = **7.2 m/min**, which is what the tool calculates and runs in 3D. The path is 12.00 m. The "In simple words" card summarises this, suggests a conveyor type, and lists what is still missing (for example the conveyor width, the electrical supply and the drive details) for Ultraplast to confirm.

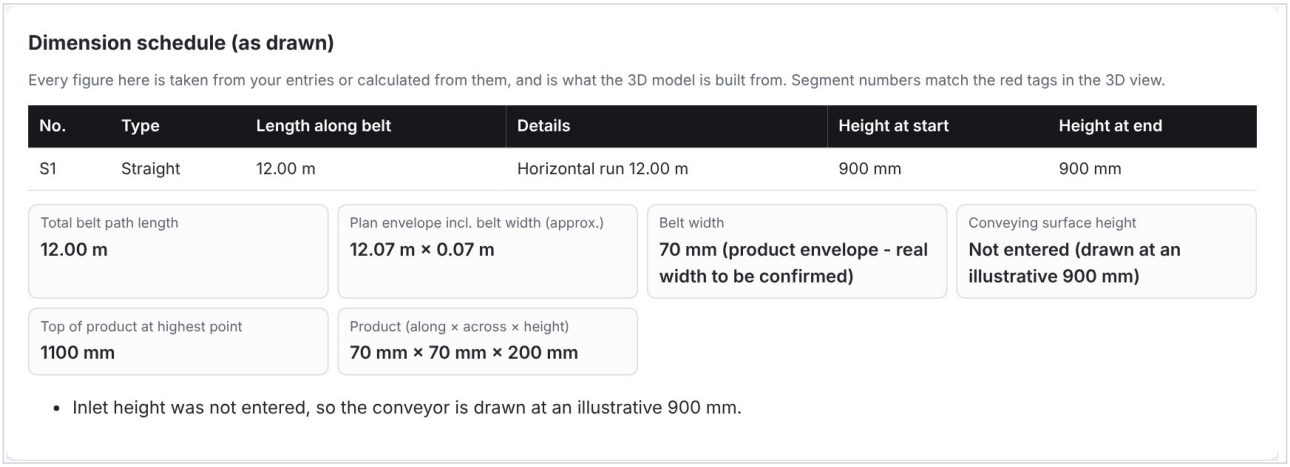


Figure 8 - The dimension schedule for the example route: every figure is taken from the entries or calculated from them.

Example 2 - A customer who knows only the conveyor length

1. Open the tool and press **Quick start: I only know the conveyor length**.
2. Type **8** in "Total conveyor route length" and press **Calculate results**.

Result: a path of 8.00 m, the 3D model, the drawing sheet, DXF files, PDF and JSON are all produced. The floor shows "Not given (optional)" and no fit is claimed. Product, throughput, width and drive are listed as missing for engineering review. Nothing is invented. The customer can press **Edit inputs** later and add details to refine the result.

Example 3 - An L-shaped line

Choose **L-shaped**, then fill the three rows: **straight 6 m**, **curve 90 degrees, radius 1000 mm, left**, **straight 4 m**. The tool calculates the curve length (1000 mm x 90 degrees = 1.571 m) and the total path of **11.571 m**. If a curve radius is too small for the chosen belt type, a red message explains the published minimum. In the 3D model, a belt speed that is too high for the curve makes products slide off.

Quick self-test (2 minutes)

Floor 20 x 6, Straight, product 100 x 100 x 250 mm, 500 g, gap 50, throughput 60 per minute, speed 10, route length 12. **Expected:** path 12.00 m, belt running at 10.00 m/min, product spacing 150 mm, minimum speed needed 9 m/min.

Part F For the Ultraplast team: setup and upkeep

This part is for staff who publish and maintain the tool. No programming is needed for the normal tasks.

F1 What is in the project folder

File or folder	Purpose
index.html	The page. Must stay at the top level of the GitHub repository.
styles.css	Colours and layout.
products.js	**Editable product and engineering data**, belt colours, friction placeholders, company details. The main file staff edit.
config.js	Settings: enquiry address, upload address, costing switch. No secrets.
calculations.js, physics.js	Calculation engine and product behaviour model.
conveyor3d.js, cad.js, schematic.js, simulation.js	3D viewer, CAD drawings and DXF writer, plan view, animation.
report.js, enquiry.js, costing.js, sheetconfig.js, app.js	PDF/JSON, enquiry sending, internal costing, Google Sheet data, the page logic.
google-sheets/	Google Apps Script that receives enquiries into a Google Sheet.
examples/	Sample project files you can load with Load JSON.
docs/	Guides: how to use, Google Sheet setup, costing, verification checklist, manufacturing readiness.
tests/	Automatic tests.

F2 Publishing on GitHub Pages (about 10 minutes)

1. Sign in to GitHub and create a **new repository** (for example `conveyor-planner`). Choose **Public** (free Pages needs a public repository). Do not tick "Add a README".
2. Choose **"uploading an existing file"** and drag in the **contents** of the project folder (not the zip itself), so that `index.html` is at the top level. Commit.
3. Go to **Settings, then Pages**. Under "Build and deployment" choose **Deploy from a branch**, branch **main**, folder **/ (root)**, and Save.
4. After a minute GitHub shows the website address (like `https://yourname.github.io/conveyor-planner/`). Open it and run the 2-minute self-test from Part E.
5. To link it from ultraplastindia.com, add a menu button to that address (or embed it in a page).

To update later: upload the changed files to the same repository and commit; the site refreshes in a minute or two.

F3 Making enquiries arrive (important)

GitHub Pages cannot receive data by itself. Until an enquiry service is connected, the tool **honestly says the enquiry was not sent** and offers an e-mail draft. To receive enquiries online:

- Create a small receiving service - the supplied **Google Apps Script** (in `google-sheets/Code.gs`) writes each enquiry as a row in a Google Sheet and can e-mail Ultraplast. Or use a form service such as Formspree or Web3Forms.
- Paste its public web address into `config.js` as `enquiry.endpoint`. For the Google script set `sendAs: 'text'` and `successField: 'ok'`.

- Send a test enquiry from the live site and check it arrives. Do the same with the internet switched off to confirm the "not sent" message appears.
- Set the WhatsApp number only after it is confirmed; until then the WhatsApp button stays disabled.

The full step-by-step is in **docs/GOOGLE_SHEETS_SETUP.md**. It was written and tested against simulated responses only - it has **not** been tested against a real Google account, so please run the test above.

F4 Editing product data

- Edit `products.js` (or the optional Google Sheet) for conveyor families, published minimum radii, width limits, colours and company details. **Only put in values that Ultraplast has verified.** If a value is empty, the tool says "Engineering review required" - it never guesses.
- Replace the **placeholder friction values** (`behaviour` section) with measured values, then set `verified: true`.
- Work through **docs/VERIFICATION_CHECKLIST.md** before public launch.

F5 Internal cost estimate (staff only)

- Open the site with `?costing=1` at the end of the address. The panel appears under the results.
- Enter your own rates per unit; quantities come from the customer's conveyor. Add wastage, overhead, margin, discount and GST. A line without a rate is flagged and the total says INCOMPLETE - it is never counted as zero.
- **Save rate card** to a file and **Load rate card** next time. Rates are never in the PDF, JSON or enquiry sent to customers.
- The switch is a convenience, not security. For complete privacy keep a private copy of the site for staff.

F6 Checking that everything still works

The project contains automatic tests (calculations, 3D geometry, drawings and DXF structure, physics, report, enquiry, costing) and browser tests (full form, downloads, falling products, length-only flow, ease of use). They need Node.js: run `npm test` and `npm run test:browser`. Ask your developer to run them after any change.

Part G Limits, FAQ, glossary and checklist

G1 Limits you should know

- **Preliminary only.** Not a final design, not a quotation, no prices.
- **Not for manufacture.** Frame, guides, legs, rollers and motor are placeholder sizes. There is no motor, tension, shaft or structural engineering, no fasteners or tolerances, and no bill of materials.
- **Behaviour model is illustrative.** Friction values are generic and unverified; collisions, vibration and real transfers are not modelled.
- **Dimensions in DXF are lines and text**, not AutoCAD dimension objects.
- **Accuracy depends on the entries.** Site measurements must be checked by Ultraplast before design.
- **Not yet verified in real conditions:** opening the DXF files in AutoCAD, a live GitHub Pages site, a real Google account, and use on a real phone. Please test these before launch.

G2 What is needed before drawings can be used for manufacturing

1. Ultraplast supplies the **real standard parts** for each conveyor type: frame profiles, legs and brackets, side guides, drive and motor options, bearings, shafts, sprockets, idlers. These are added to the product data and replace the placeholders.
2. A real **bill of materials** and standard detail views are added (frame section, drive end, idler end, leg and foot).
3. Engineers complete and record the **calculations** (power, tension, structure).
4. A **sign-off step**: an Ultraplast engineer reviews and approves each drawing before it is released. Until then, drawings stay "not for manufacture".
5. The DXF files are tested in real AutoCAD and any problem is fixed.

G3 Frequently asked questions

Question	Answer
Do I have to fill everything?	No. Only the conveyor length is needed. Leave anything else blank or press Skip this step.
I do not know the floor size. Can I still use it?	Yes. Leave the floor boxes empty. The conveyor is drawn on its own and the fit on site is not checked.
Does the tool give a price?	No. It never shows prices. Ultraplast sends a quotation after engineering review.
Is my data stored or sent?	No. Everything stays in your browser. Nothing is sent until you press Submit, and nothing is saved when you close the page.
Can I save my work and continue later?	Yes: download the JSON file, and open it later with Load JSON.
The 3D picture shows a belt width I did not enter.	When no width is entered, an illustrative width is drawn and the page says "not entered". Ultraplast confirms the real width.
Why do products fall off in the 3D model?	The belt speed is higher than the highest safe speed for that curve and belt, or the grip is low. Lower the speed or use a larger curve radius.
Why does it say "Engineering review required"?	The item cannot be safely calculated from the information given (for example motor power). An Ultraplast engineer will decide it.
Can I open the drawing in AutoCAD?	Yes - download the 2D or 3D DXF and use OPEN. It is full size in millimetres. Type ZOOM then E if the drawing looks empty.
Can the drawing go to the workshop?	No. It is indicative. Only engineer-approved drawings may be used for manufacturing.
The PDF or download did not start.	Allow downloads for this website in your browser, or try another browser. On a phone, check the Downloads folder.
The enquiry says it was NOT sent.	Online sending is not connected yet or the connection failed. Use "E-mail Ultraplast (draft)" and attach the PDF, or call Ultraplast.
A red message says the route does not fit.	Shorten the route, change the layout, or move the start point. This is only checked when a floor size is entered.

G4 Glossary

Term	Meaning
Throughput	How many products pass the end of the conveyor per minute or hour.
Pitch	Product length along the belt plus the gap to the next product.
Gap	Free space between one product and the next.
Lane	A row of products running side by side across the belt.
Inlet / discharge	The place where products get on / off the conveyor.
Centre-line	The line down the middle of the belt; route lengths are measured along it.
Radius (curve)	Distance from the curve's centre to the belt centre-line. A small radius is a tight bend.
Incline / decline	A section that rises / falls. Run = horizontal distance; rise = height change.
Footprint	The floor space the route covers.
Transfer	Where products pass from one conveyor to the next.
Friction (mu)	How well the belt grips the product. Low = slides easily.
Centre of gravity	The balance point of a product. A high one tips over more easily.
Accumulation	Products queue up and touch each other on the belt.
Washdown	Cleaning with water, so the conveyor must resist it.
DXF	A drawing file format that AutoCAD and many CAD programs open.
SVG	A drawing format that opens in a web browser and can be printed to PDF.
JSON	A plain data file holding all your entries and results; used to save and re-open projects.
Indicative	Shown only to make the picture look right; not an Ultraplast specification.
Engineering review	A decision that must be made by an Ultraplast engineer.

G5 Pre-launch checklist for staff

	Check
[ ]	Product data in products.js reviewed; every value verified by engineering.
[ ]	Friction and centre-of-gravity placeholders replaced with measured values (verified: true).
[ ]	Enquiry service connected; test enquiry received; "not sent" message seen when offline.
[ ]	WhatsApp number confirmed (or left disabled).
[ ]	Company phone numbers and e-mail on the page are correct.
[ ]	2D and 3D DXF opened in AutoCAD; layers and dimensions look right.
[ ]	Tried on a real phone (form, 3D view, downloads).
[ ]	Run the 2-minute self-test on the live website.
[ ]	Everyone understands: preliminary only, no prices, drawings not for manufacture.
[ ]	Link added on ultraplastindia.com.

Need help?

Contact Ultraplast through the details shown at the bottom of the tool page, or write to the address configured in the project. For changes to the tool itself, give your developer this guide and the project's **docs** folder.